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EDUCATION			
Program	Institute	CGPA/%	Year
B.E. in Electronics & Communication Engineering	Sardar Vallabhai Patel Institute of Technology	7.98	2023
ClassXII(CBSE)	Podar World School ,Sherkhi	74.4%	2019
ClassX(CBSE)	New Era Senior Secondary School,Vadodara	7.8	2016
PROFESSIONALEXPERIENCE			
Embedded Techno solutions Internship (July'25-February'26)	Embedded Systems and PCB Designing		
	➤ Developed firmware for both 8-bit (8051) and 32-bit (ARM7 LPC2148) platforms, gaining a comprehensive understanding of memory mapping, registers, and clock configurations. ➤ Programmed and interfaced essential hardware modules including LCDs, Keypads, ADCs, UART (Serial Communication), and Timers/Counters using Embedded C. ➤ Designed and validated complex circuit schematics using Proteus Design Suite, ensuring functional accuracy before hardware implementation. ➤ Implemented hardware and software interrupts to manage real-time tasks and optimize CPU efficiency on the LPC2148. ➤ Acquired foundational knowledge of Single Board Computers (SBCs), configuring the Raspbian OS and managing basic GPIO operations. ➤ Developed entry-level Python scripts to automate hardware tasks and read digital sensor data on the Raspberry Pi platform. ➤ End-to-End PCB Development in Altium Designer, from schematic capture and component footprint creation to multi-layer routing and the generation of manufacturing outputs (Gerber and NC Drill files). ➤ Engineered functional hardware layouts for an AC-to-DC Power Converter and a custom Arduino Nano system, implementing design principles for power regulation, signal integrity, proper grounding, power plane isolation to minimize noise and electromagnetic interface in mixed signal circuits and compact component placement.		
Sofcon India Pvt. Ltd Trainee (April'24-August'24)	Industrial Automation and Control Systems		
	➤ Gained hands-on in developing ladder logic and functional block diagrams for Allen-Bradley Mircrologix1000 and Siemens S-1200, automating industrial sequences. ➤ Designed intuitive industrial interfaces using SIMATIC HMI (KTP400 Basic), focusing on real-time data monitoring, alarm management, and user-friendly control dashboards. ➤ Configured and tuned Allen-Bradley Power Flex 40 VFD to optimize motor performance, speed control, and energy efficiency in automated systems. ➤ Implemented communication protocols (Profinet/Ethernet/IP) to establish seamless data exchange between PLCs, HMIs, and field devices. ➤ Studied and applied concepts of Smart Manufacturing, including IoT integration, remote monitoring, and data acquisition for predictive maintenance frameworks.		
L&T Technology Services Limited (Dept.-Digital Manufacturing Services) Internship (January'23–April'23)	Simulations, Scripting, Commissioning and Automation		
	➤ Learned to design and implement SCADA systems, including basic user interface and configuring data sources. ➤ Managed and visualized real-time and historical data from industrial Processes by Ignition Software ➤ Acquired skills in integrating Ignition software with various industrial protocols and databases for seamless dataflow and control. ➤ Basic Understanding in diagnosing and resolving issues with the SCADA system to ensure optimal performance.		
PROJECTS			
IOT Enabled Real Time Soil Monitoring System	Guide :Hiren J Modha ,Mini Project		
	• Developed an IOT based Soil Monitoring system which measures soil pH Level, Temperature Humidity of Soil. • The main objective of this project was to create a system that will allow farmer and plant grower to keep monitoring the soil condition and impact of pH on plant growth.		

	Technologies Used: C programming, Thing Speak, Arduino IDE Software		
Smart Attendance system using 8051 microcontroller	Guide: Tushar Kulkarni ,Course project		
	- Designed and implemented a password-protected attendance system using Embedded C and the 8051 microcontroller architecture. - Developed a custom scanning algorithm for a 4x4 Matrix Keypad to capture user input efficiently with debouncing logic. - Interfaced a 16x2 LCD module to provide real-time visual feedback and user prompts using 8-bit parallel communication. - Simulated and validated the complete hardware circuitry in Proteus to ensure logic accuracy.		
COURSEWORK AND SKILLS			
Analog Electronics	Signal Systems and Networks	Embedded System Design	
Control Systems	Electronic Devices and Circuits	Communication System	
Industrial Automation	RF and Satellite Communication	Electronic product Design	
Programming	C,C++,Embedded C,Python	Software	Proteus, Keil, Altium Designer, TIA Portal, RS Logix 500
Libraries	Pandas, RPi.GPIO, Reg51.h, FreeRTOS	Project Management	Msoffice,Canva
Courses & Certifications			
IIT Madras Pravartak (Swayam Plus)	Comprehensive Electronics and Embedded Systems: From Basic to Applications.		
Fasal Agritech	Industry 4.0, IOT & Product design.		
Udemy (Online)	Learn Scada from Scratch-Design, Program and Interface		
IT SOULS	Programming in C		